

MATT WIENS

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SKILLS

Programming Languages: Python, TypeScript, C++, Java, R, Lua

Frameworks and Tools: React, Express, Django, SQL, Nginx, D3.js, Docker, Ansible

Proficiencies: Full-stack, CI/CD, cloud, containerization, data science, scientific computing

PROJECTS

Super Mario 64 Route Optimizer | github.com/mwiens91/sm64-route-optimizer 2025
Video Game Route Optimization Tool for Speedrunning

- Designed and implemented an optimization algorithm for a novel constrained selection problem
- Translated gameplay mechanics into formal constraint structures, including prerequisite relationships, mutually exclusive action paths, and global resource-based access conditions
- Built a configurable Python tool that generates optimized routes from user-defined inputs

EXPERIENCE

University of Waterloo, Waterloo, Canada Sep 2020–Sep 2023
Graduate Teaching Assistant

- Led weekly tutorials for 100 to 300-level mathematics courses, teaching 25–40 students per session
- Developed and delivered short lectures and provided individualized student support
- Collaborated with instructors to develop marking schemes for assignments and exams

BC Cancer: Shah Lab, Vancouver, Canada May 2018–Dec 2018
Bioinformatics Software Developer (Co-op Position)

- Developed and maintained two large Django-based genomic databases in a team environment
- Implemented analysis workflows in R for single-cell RNA-seq data to perform cell type classification
- Independently built **Saltant** (github.com/saltant-org/saltant), a distributed task orchestration system for running containerized workflows across on-prem and cloud servers
 - Features a web interface, REST API, Python SDK, and Unix CLI; developed with Django, Celery, and RabbitMQ; includes full documentation and high-coverage test suite
 - Served as a core component of BC Cancer's Molecular Oncology dry lab infrastructure for 7 years with minimal maintenance

EDUCATION

The University of British Columbia, Vancouver, Canada Sep 2025–present
Bachelor of Computer Science (BCS)

- GPA 98.9/100

University of Waterloo, Waterloo, Canada Sep 2020–Sep 2023
Master of Mathematics in Applied Mathematics (partial completion)

- NSERC CGS M scholarship (\$17,500 over one year)
- Received 4 scholarships and awards totalling \$34,500
- GPA 90.67/100

Simon Fraser University, Burnaby, Canada Sep 2015–Aug 2020
Bachelor of Science (Honours First Class with Distinction)

- Honours in Applied Mathematics, Minor in Physics
- Undergraduate Dean's Medal, Faculty of Science
- Received 18 scholarships and awards totalling \$23,935
- GPA 4.21/4.33